Text Classification using Sparse Features and Contextual Embeddings

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Course Project Report

1 Introduction and Dataset Used

- This project focuses on text classification using both sparse representations and contextual embeddings.
- The AG News dataset is used, consisting of news articles categorized into four classes (World, Sports, Business, Sci/Tech).
- The objective is to compare traditional methods (BoW, TF-IDF) with modern embedding techniques (BERT, DistilBERT).

2 Dataset Description and EDA

- The dataset contains 10,000 samples (8000 training and 2000 testing) across 4 balanced classes.
- Text data varies in length and vocabulary richness, requiring preprocessing and normalization.
- EDA helps understand class distribution, word frequency, and text patterns.

3 Methodology

- Text preprocessing includes lowercasing, stopword removal, and noise filtering.
- Feature extraction is performed using Bag of Words and TF-IDF (bigram). TF-IDF with bigrams was used to capture both term importance and local context.
- Contextual embeddings were generated using pre-trained transformer models (DistilBERT and BERT).
- Dimensionality reduction techniques such as Truncated SVD and PCA are applied.

4 Experiments

- Multiple models including Logistic Regression, Linear SVM, KNN, and KMeans were evaluated.
- Experiments were conducted on both sparse features and embedding-based features.
- Hyperparameters such as number of neighbors (KNN) and clusters (KMeans) were tuned.

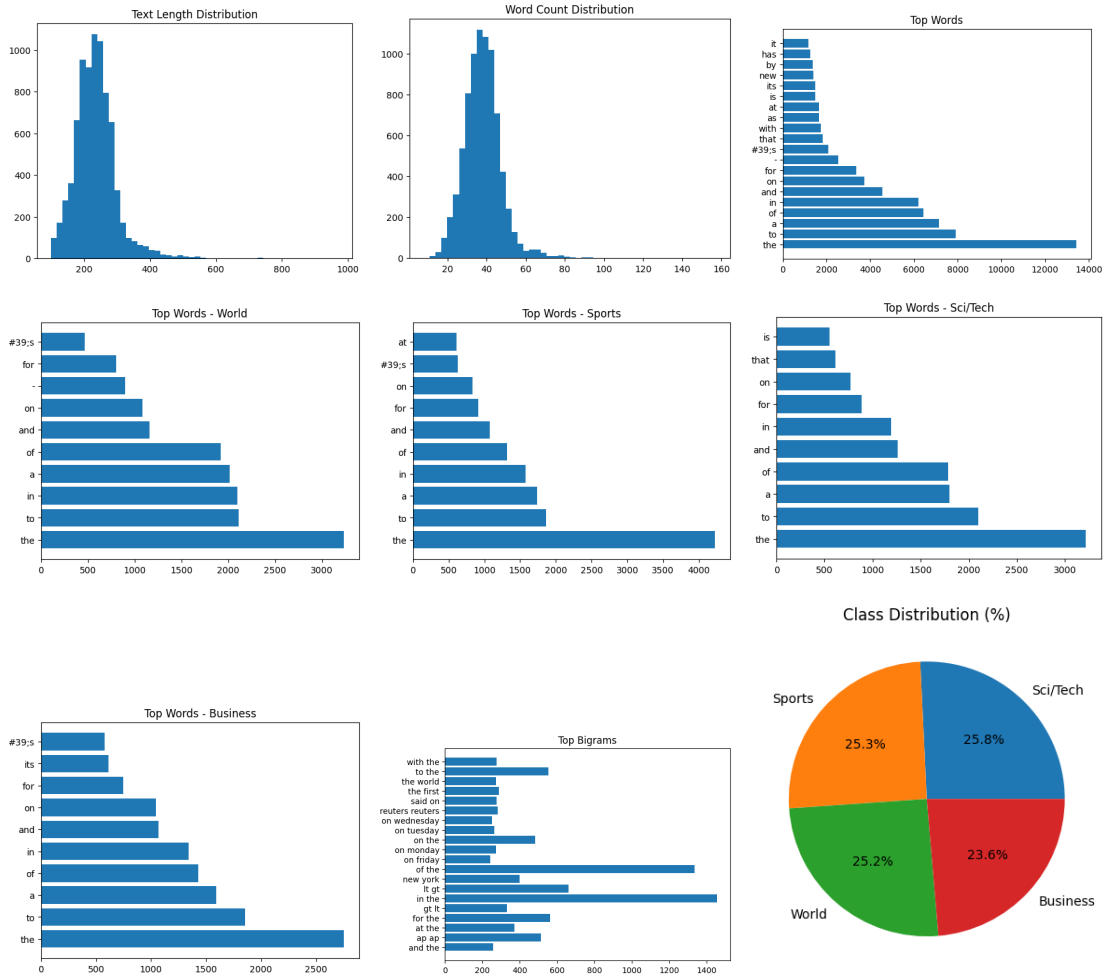


Figure 1: Exploratory Data Analysis (EDA) Visualizations

5 Preprocessing and Vocabulary Pruning

Steps applied:

- Lowercasing
- Removing special characters
- Stopword removal
- Removing short words (length ≤ 2)
- Removing numeric tokens

This reduces noise and improves feature quality.

6 Sparse Features

6.1 Bag of Words (Unigram)

Vocabulary size:

- BoW Vocabulary: 5000

6.2 TF-IDF (Bigram)

- TF-IDF Vocabulary: 5000
- Bigram model used

7 Dimensionality Reduction

7.1 Truncated SVD

Observation: SVD reduces dimensionality but retains limited variance, indicating information loss.

7.2 PCA on TF-IDF

- Components: 3187

Observation: PCA reduces dimensionality significantly while preserving structure.

8 Model Performance (Sparse Features)

Model	Feature Type	Accuracy	Macro F1	Recall	Precision
Logistic Regression	BoW	**0.8465**	**0.8474**	**0.8482**	**0.8472**
Linear SVM	BoW	0.8085	0.8095	0.8102	0.8092
KNN	BoW	0.2790	0.2755	0.2803	0.2769
KMeans	BoW	0.2980	0.2468	0.2975	0.3547
Logistic Regression	TF-IDF	**0.7575**	**0.7632**	**0.7590**	**0.7851**
Linear SVM	TF-IDF	0.7465	0.7518	0.7480	0.7722
KNN	TF-IDF	0.4960	0.4840	0.5008	0.5400
KMeans	TF-IDF	0.2420	0.0983	0.2505	0.3104

9 Visualizations (Sparse Features)

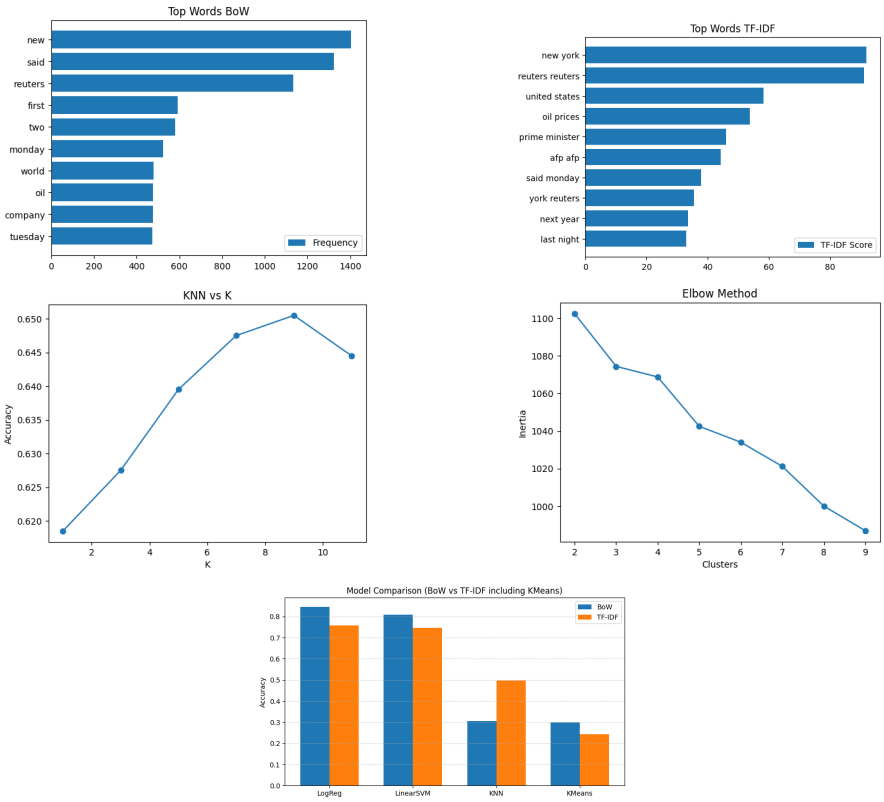


Figure 2: Sparse Features Visualization

10 KNN Analysis

- Best $k = 9$

Observation:

- Small $k \rightarrow$ noisy
- Large $k \rightarrow$ oversmoothing

11 KMeans Analysis

Elbow Method:

- Optimal Clusters $\rightarrow k = 4$

12 PCA on Contextual Embeddings

We used:

12.1 DistilBERT

- PCA Components: 247
- Explained Variance: 0.9501

12.2 BERT

- PCA Components: 383
- Explained Variance: 0.9501

Observation:

- PCA retains 95% variance
- Significant dimensionality reduction

13 Embedding Model Results

13.1 DistilBERT

Model	Accuracy	Macro F1	Recall	Precision
LogReg	0.8755	0.8767	0.8770	0.8766
LinearSVM	**0.8825**	**0.8836**	**0.8839**	**0.8836**
KNN	0.8705	0.8718	0.8720	0.8723
KMeans	0.6550	0.6739	0.6585	0.7108

13.2 BERT

Model	Accuracy	Macro F1	Recall	Precision
LogReg	**0.8770**	**0.8780**	**0.8782**	**0.8780**
LinearSVM	0.8755	0.8766	0.8766	0.8773
KNN	0.8315	0.8330	0.8328	0.8343
KMeans	0.5490	0.5649	0.5487	0.6160

14 Embedding Visualizations

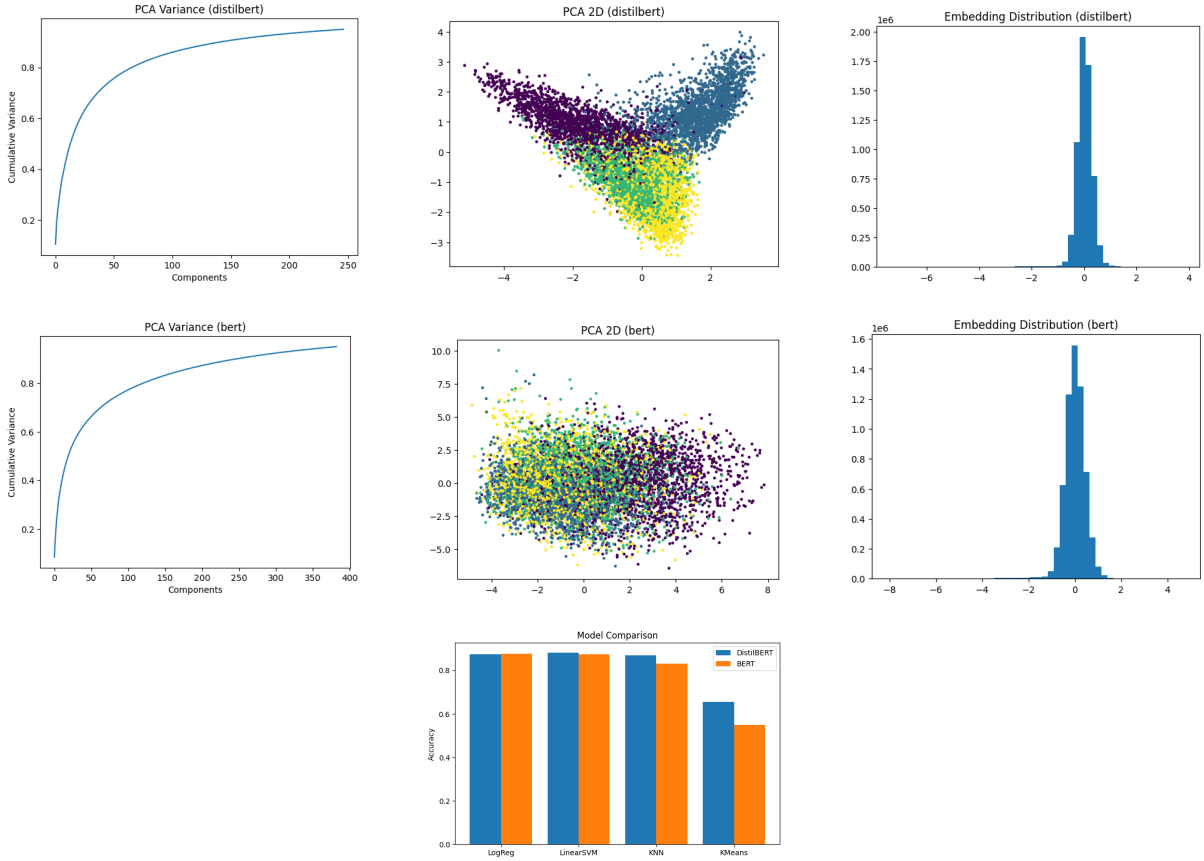


Figure 3: Embedding Visualization

15 Results

- Across all experiments, Logistic Regression consistently achieved the most stable and highest overall performance, particularly on sparse feature and BERT embeddings.
- Linear SVM showed competitive performance and achieved the best results on DistilBERT embeddings, suggesting that margin-based classifiers benefit from well-structured dense representations.
- KNN exhibited moderate performance, performing poorly on sparse high-dimensional

data due to the curse of dimensionality, but improving significantly on embedding-based features where distance metrics become more meaningful.

- KMeans consistently showed the lowest performance, as it is an unsupervised clustering method and lacks access to class labels, making it unsuitable for achieving high classification accuracy in this setting.

16 Discussion

- Although TF-IDF typically outperforms BoW, in our case BoW achieved higher accuracy. This may be due to aggressive preprocessing and limited dataset size, where frequency-based features retained more useful information than weighted representations.
- DistilBERT performs comparably to BERT because it is a compressed version of BERT that retains most of its knowledge while being more efficient.
- KNN performs poorly on sparse data due to high dimensionality and sparsity, but performs significantly better on embeddings where data is dense and distance metrics become meaningful.

17 Conclusion and Learnings

- Dense embeddings outperform sparse features as they capture semantic context, whereas sparse methods rely only on word frequency.
- Logistic Regression performed most consistently and best overall across feature types, with Linear SVM achieving the highest performance on DistilBERT embeddings.
- In our experiments, Bag of Words slightly outperformed TF-IDF, likely due to dataset size and preprocessing, although TF-IDF generally performs better on larger datasets.
- DistilBERT offers the best trade-off between efficiency and performance, achieving results comparable to BERT with reduced computational requirements.
- Dimensionality reduction techniques such as PCA are essential for handling high-dimensional text data while preserving most of the important variance.